ATP synthase, also known as Complex V, produces the majority of the ATP in eukaryotic cells. It is a transmembrane protein encoded by both nuclear and mitochondrial genes in humans, and is found in the inner mitochondrial membrane. Complex V consists of over 20 subunits, including three sets of heterodimers arranged in a circle in the mitochondrial matrix. Each dimer contains an α subunit and a β subunit, and each β subunit exists in one of three conformations: one that binds ATP, one that binds ADP and inorganic phosphate, and one that binds neither.

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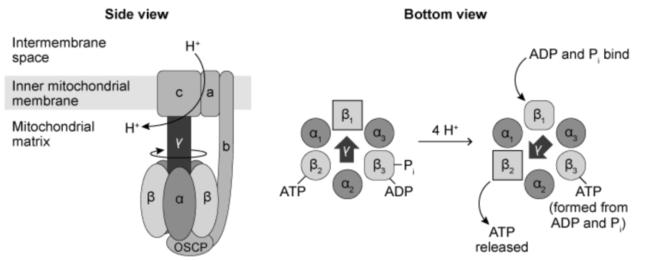


Figure 1 Depictions of ATP synthase viewed from the side (left) and the bottom (right)

Researchers visualized the rotation of the γ subunit by attaching it to a fluorescently labeled actin filament. The observed rotation over time is shown in Figure 2.

Figure 2 Counterclockwise rotation of the γ subunit of ATP synthase (bottom view) over time in seconds as visualized by a fluorescent label

An L156P mutation in subunit A of Complex V is linked to neuropathy, ataxia, and retinal degeneration. Although the exact molecular bases of these pathological states are unknown, the following hypotheses are currently under study.

Hypothesis 1

L156P Complex V functions normally but is less abundant than wild-type Complex V.

Hypothesis 2

The mutation partially decouples proton flow from ATP synthesis.

Hypothesis 3

The mutation leads to increased oxidative stress from the ETC, resulting in apoptosis.

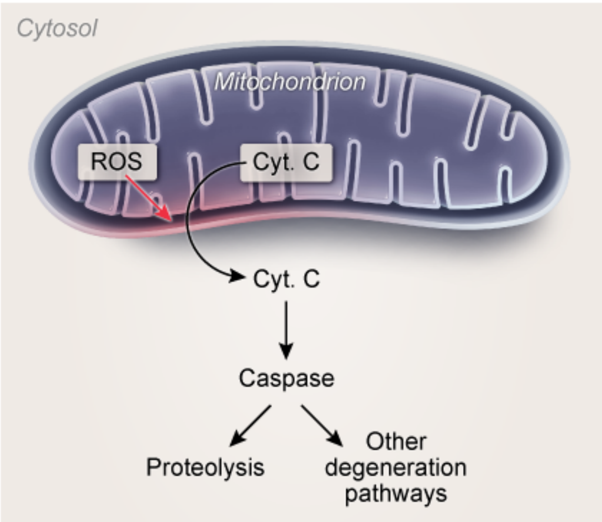
All of the following could confirm apoptosis due to increased oxidative stress in cells with an L156P mutation in Complex V EXCEPT:

- ✔ A. inhibition of caspase activity. [54%]
- B. increased cytosolic cytochrome C. [26%]
- C. activation of proteolysis. [13%]
- D. increased mitochondrial superoxide concentration. [5%]

Correct
53% answered correctly

Explanation:

Apoptosis induced by oxidative stress



Oxidative stress occurs when cells have an overabundance of **reactive oxygen species** (ROS), which can cause damage when they react with cell membranes and other biological molecules. The ETC in the mitochondria produces ROS such as superoxide, hydrogen peroxide, and hydroxyl radicals when it fails to fully reduce oxygen to water. Normal levels of ROS (about 4% of oxygen consumed) are removed by enzymes such as peroxidase and superoxide dismutase. However, defects in Complex V and other ETC complexes can cause oxidative stress by increasing the mitochondrial

concentrations of superoxide and other ROS beyond the ability of the cell to control **(Choice D)**.

Apoptosis is the controlled, **programmed death** of cells. It occurs in response to a variety of circumstances, including oxidative stress as well as DNA damage and certain developmental events. Apoptosis is initiated when the mitochondrial membrane is permeabilized in response to these environmental cues. This allows the release of cytochrome C from the mitochondria, resulting in **increased cytosolic cytochrome C (Choice B)**. Cytochrome C induces **proteolysis (Choice C)** and other degradative pathways by **activating caspase** proteases (inactive caspase is found in *healthy* cells).

Educational objective:
Apoptosis (programmed cell death) can be caused by certain developmental events, DNA damage, or reactive oxygen species. It is induced when cytochrome C is allowed to leave the mitochondria and enter the cytosol, where it activates caspase. Caspase in turn activates several degradative pathways such as proteolysis.

Time spent:0 sec
Subject:
Biochemistry

QID:401501
System:
Metabolic Reactions

ATP synthase, also known as Complex V, produces the majority of the ATP in eukaryotic cells. It is a transmembrane protein encoded by both nuclear and mitochondrial genes in humans, and is found in the inner mitochondrial membrane. Complex V consists of over 20 subunits, including three sets of heterodimers arranged in a circle in the mitochondrial matrix. Each dimer contains an α subunit and a β subunit, and each β subunit exists in one of three conformations: one that binds ATP, one that binds ADP and inorganic phosphate, and one that binds neither.

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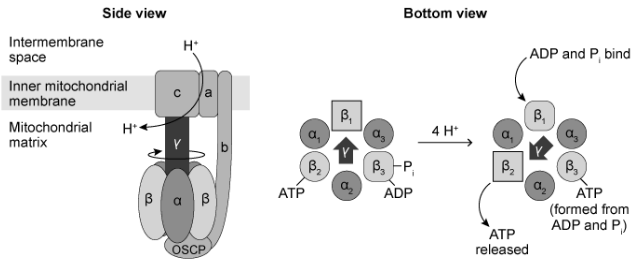


Figure 1 Depictions of ATP synthase viewed from the side (left) and the bottom (right)

Researchers visualized the rotation of the γ subunit by attaching it to a fluorescently labeled actin filament. The observed rotation over time is shown in Figure 2.

Figure 2 Counterclockwise rotation of the γ subunit of ATP synthase (bottom view) over time in seconds as visualized by a fluorescent label

An L156P mutation in subunit A of Complex V is linked to neuropathy, ataxia, and retinal degeneration. Although the exact molecular bases of these pathological states are unknown, the following hypotheses are currently under study.

Hypothesis 1

L156P Complex V functions normally but is less abundant than wild-type Complex V.

Hypothesis 2

The mutation partially decouples proton flow from ATP synthesis.

Hypothesis 3

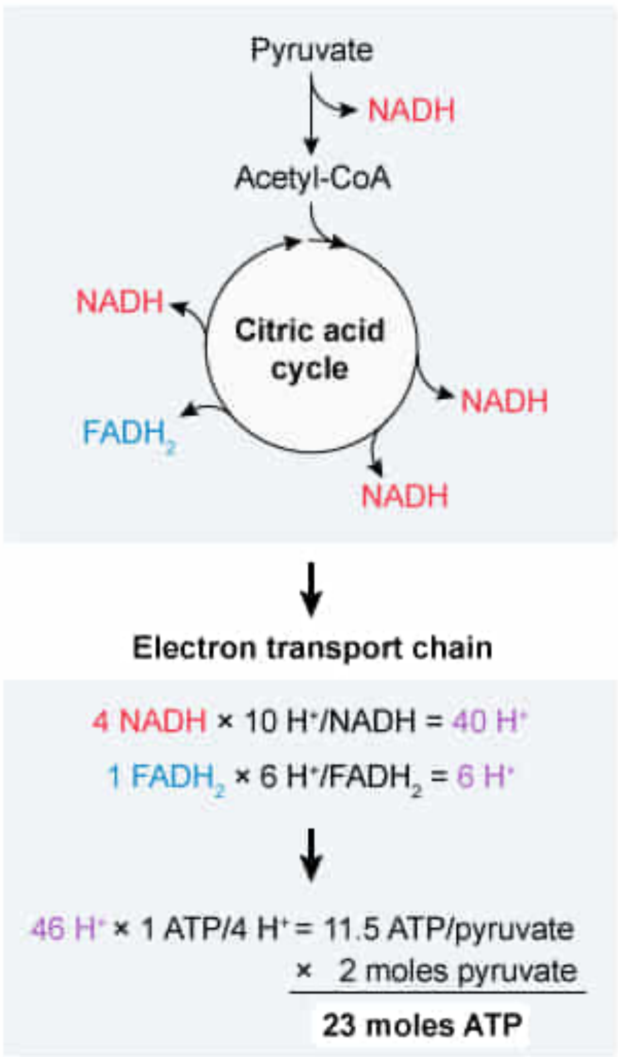
The mutation leads to increased oxidative stress from the ETC, resulting in apoptosis.

When pyruvate enters the mitochondria, it is converted to acetyl-CoA and further oxidized in a series of reactions that produces four NADH molecules and one $FADH_2$ molecule. If all the NADH and $FADH_2$ produced by 2 moles of pyruvate digestion in the mitochondria enter the ETC, how many moles of ATP can wild-type Complex V produce?

- A. 18 [9%]
- B. 23 [66%]
- C. 28 [11%]
- D. 32 [13%]

Correct
66% answered correctly

Explanation:



Pyruvate enters the mitochondria, where it is oxidized to three carbon dioxide molecules in a series of enzymatic steps. Each pyruvate molecule produces **four NADH molecules** (one when pyruvate is converted to acetyl-CoA and three more in the citric acid cycle) and **one $FADH_2$ molecule**. These molecules can enter the ETC, where the energy they release pumps protons from the mitochondrial matrix into the intermembrane space. ATP is generated when these protons flow back into the matrix through Complex V in a process called oxidative phosphorylation. The ATP generated by Complex V is proportional to the amount of NADH and $FADH_2$ generated.

According to the passage, **10 protons** are pumped into the intermembrane space **per NADH** molecule that enters the ETC, and **6 protons** are pumped **per $FADH_2$** molecule. Therefore, each pyruvate can cause the pumping of **46 protons** (ie, 40 from the four NADH molecules and 6 from $FADH_2$). **The passage also states** that Complex V produces **3 ATP for every 12 protons** (or 1 ATP per 4 protons) that flow through it; therefore, it produces 46/4 = 11.5 ATP molecules for every pyruvate molecule, or 11.5 moles of ATP for every mole of pyruvate. Because the question asks for the ATP yield of **2 moles of pyruvate**, this result must be doubled, giving **23 moles of ATP** for 2 moles of mitochondrial pyruvate.

(Choice A) The NADH produced when pyruvate is converted to acetyl-CoA must be accounted for in the calculation. Leaving it out will yield an answer of 18 moles of ATP.

(Choice C) Glycolysis produces two NADH molecules during the conversion of glucose to pyruvate; however, this happens before digestion of pyruvate in the mitochondria. Counting them in the calculation yields 28 moles of ATP.

(Choice D) The generally accepted maximum yield from 1 mole of glucose is 32 moles of ATP. However, this number includes ATP that is generated by substrate-level phosphorylation during glycolysis and the citric acid cycle, which do not involve Complex V.

Educational objective:
The ATP yield of oxidative phosphorylation is directly proportional to the amount of NADH and $FADH_2$ that enters the electron transport chain: 1.5 ATP per $FADH_2$ and 2.5 ATP per NADH.

Time spent:0 sec
Subject:
Biochemistry

QID:401502
System:
Metabolic Reactions

ATP synthase, also known as Complex V, produces the majority of the ATP in eukaryotic cells. It is a transmembrane protein encoded by both nuclear and mitochondrial genes in humans, and is found in the inner mitochondrial membrane. Complex V consists of over 20 subunits, including three sets of heterodimers arranged in a circle in the mitochondrial matrix. Each dimer contains an α subunit and a β subunit, and each β subunit exists in one of three conformations: one that binds ATP, one that binds ADP and inorganic phosphate, and one that binds neither.

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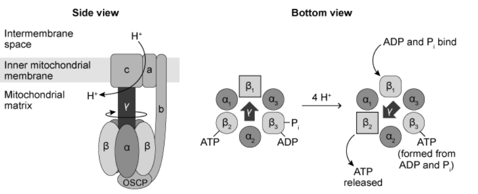


Figure 1 Depictions of ATP synthase viewed from the side (left) and the bottom (right)

Researchers visualized the rotation of the γ subunit by attaching it to a fluorescently labeled actin filament. The observed rotation over time is shown in Figure 2.

Figure 2 Counterclockwise rotation of the γ subunit of ATP synthase (bottom view) over time in seconds as visualized by a fluorescent label

An L156P mutation in subunit A of Complex V is linked to neuropathy, ataxia, and retinal degeneration. Although the exact molecular bases of these pathological states are unknown, the following hypotheses are currently under study.

Hypothesis 1

L156P Complex V functions normally but is less abundant than wild-type Complex V.

Hypothesis 2

The mutation partially decouples proton flow from ATP synthesis.

Hypothesis 3

The mutation leads to increased oxidative stress from the ETC, resulting in apoptosis.

If the pathological effects observed in patients with the L156P mutation are caused by decreased Complex V activity, the effects could be best counteracted by a drug that increases which of the following?

✔ A. Malate-aspartate shuttle activity to transfer cytosolic NADH to the mitochondria [66%]

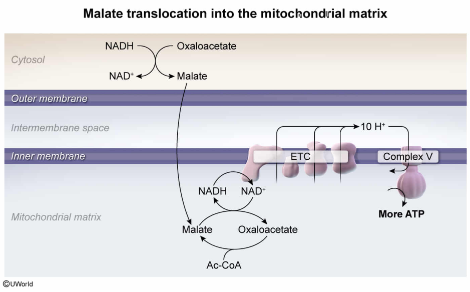
B. Potassium concentration in the mitochondrial matrix to decrease the charge gradient [12%]

C. The pH of the intermembrane space to equalize it with the pH of the matrix [10%]

D. Electron transfer from NADH to FAD via reduction of glycerol 3-phosphate to DHAP [11%]

Correct
67% answered correctly

Explanation:



The question indicates that an L156P mutation causes a decrease (though not a complete shutdown) of Complex V activity. The **activity of Complex V** depends on the **number of protons available** to flow through it. When more protons are available to flow through the complex, its activity increases. Therefore, a decrease in activity could be counteracted by an increase in proton concentration in the intermembrane space, which is dependent on the number of NADH and $FADH_2$ molecules that enter the ETC.

Glycolysis produces two cytosolic NADH molecules per glucose consumed, but they are unable to enter the mitochondrial matrix directly. Instead, cytosolic NADH must transfer its electrons to the matrix through an intermediate. This can be achieved most efficiently by reducing oxaloacetate to malate, which can enter the mitochondrial matrix (**malate-aspartate shuttle**). In the matrix, malate is oxidized back to oxaloacetate in the last step of the citric acid cycle, generating mitochondrial NADH in the process. This process effectively transfers cytosolic NADH to the mitochondria, increasing the total proton output by the ETC. Therefore, increased malate-aspartate shuttle activity would help counteract decreased Complex V activity.

(Choice B) The energy released when protons flow through Complex V is partially due to the difference in charge across the inner membrane. Decreasing the charge difference by adding potassium to the matrix would cause a *reduction* in ATP synthesis.

(Choice C) ATP synthase is driven by protons flowing from a high concentration in the intermembrane space toward a low concentration in the matrix. Equalizing the pH of the intermembrane space and the mitochondria by raising the pH of the intermembrane space would destroy the proton gradient and stop ATP synthesis.

(Choice D) The glycerol 3-phosphate shuttle effectively converts cytosolic NADH to mitochondrial $FADH_2$ by *oxidizing* glycerol 3-phosphate to DHAP, not by reducing it. In addition, $FADH_2$ pumps fewer protons than NADH does, so this system is less efficient than malate translocation at stimulating Complex V activity.

Educational objective:
Complex V activity depends on proton availability. Proton concentration in the intermembrane space can be increased by increasing the number of NADH molecules that enter the electron transport chain. NADH from glycolysis can indirectly enter the mitochondrial matrix by first passing its electrons to oxaloacetate, forming malate.

Time spent:0 sec
Subject:
Biochemistry

QID:401503
System:
Metabolic Reactions

ATP synthase, also known as Complex V, produces the majority of the ATP in eukaryotic cells. It is a transmembrane protein encoded by both nuclear and mitochondrial genes in humans, and is found in the inner mitochondrial membrane. Complex V consists of over 20 subunits, including three sets of heterodimers arranged in a circle in the mitochondrial matrix. Each dimer contains an α subunit and a β subunit, and each β subunit exists in one of three conformations: one that binds ATP, one that binds ADP and inorganic phosphate, and one that binds neither.

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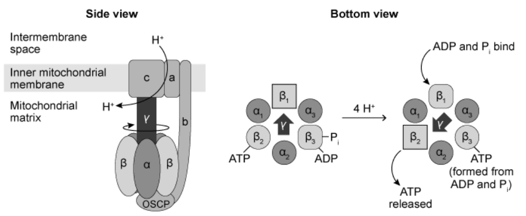

Figure 1 Depictions of ATP synthase viewed from the side (left) and the bottom (right)

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Hypothesis 2

The mutation partially decouples proton flow from ATP synthesis.

Hypothesis 3

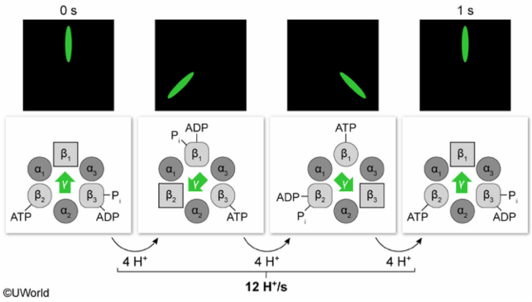
The mutation leads to increased oxidative stress from the ETC, resulting in apoptosis.

Based on the information in the passage and Figures 1 and 2, approximately how many protons flow through wild-type Complex V per second under the experimental conditions?

- A. 4 [25%]
- B. 6 [15%]
- ✔ C. 12 [54%]
- D. 24 [4%]

Correct
53% answered correctly

Explanation:



In Figure 2, researchers labeled the γ subunit with a fluorescent actin filament. The counterclockwise rotation of the filament corresponds to the rotation of the γ subunit and shows that under the given set of experimental conditions, **one complete rotation takes 1 second**.

The passage and Figure 1 indicate that 4 protons must flow through Complex V to synthesize 1 ATP molecule, and that one complete rotation yields 3 ATP molecules as 12 protons flow through the complex. Because one complete rotation occurs each second, the **rate of proton flow** in each wild-type Complex V unit must be **12 protons per second**.

(Choice A) 4 protons pass into the matrix for each $\frac{1}{3}$ of a complete revolution of Complex V. A complete revolution takes 1 second, so 4 protons flow through in $\frac{1}{3}$ of a second.

(Choice B) 6 protons would be correct if one complete rotation took 2 seconds. However, Figure 2 shows that *two* complete rotations occur in 2 seconds.

(Choice D) 24 protons would be correct if *two* rotations occurred per second instead of one. However, the counterclockwise rotation of the filament corresponds to only *one* complete rotation per second.

Educational objective:

The rate of a chemical reaction or process can be calculated by dividing the total output by the time it takes to achieve that output.

Time spent:0 sec
Subject:
Biochemistry

QID:401504
System:
Metabolic Reactions

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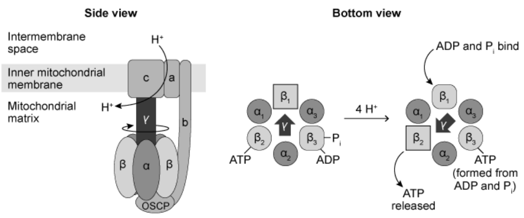

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Figure 2 Counterclockwise rotation of the γ subunit of ATP synthase (bottom view) over time in seconds as visualized by a fluorescent label

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Hypothesis 1

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Hypothesis 2

The mutation partially decouples proton flow from ATP synthesis.

Hypothesis 3

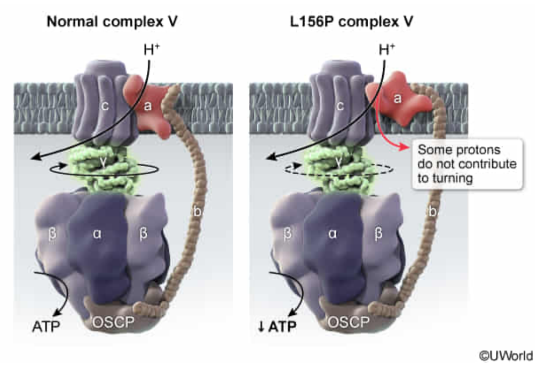
The mutation leads to increased oxidative stress from the ETC, resulting in apoptosis.

A decrease in the observed rate of ATP synthase rotation in mitochondria isolated from L156P cells relative to that observed in mitochondria isolated from wild-type cells would support which of the hypotheses given in the passage?

- A. Hypothesis 1 only [5%]
- B. Hypothesis 2 only [50%]
- C. Hypotheses 1 and 2 only [25%]
- D. Hypotheses 2 and 3 only [18%]

Correct
50% answered correctly

Explanation:



Two **chemical processes** are said to be **coupled** when the energy required for one is provided by the other. In the case of ATP synthesis by Complex V, the energy required for the phosphorylation of ADP to form ATP is provided by the flow of protons from high concentration to low concentration through the complex. The energy is transferred to the complex as the proton flow causes the γ subunit to spin, normally requiring four protons to form ATP as the γ subunit spins from one β subunit to the next.

When two processes that are normally coupled become **decoupled**, the process that provides the energy continues but the process driven by that energy slows or stops, as the two have become disconnected. In this case, the same rate of proton flow into the matrix results in **slower spinning** of the γ subunit and reduced ATP synthesis. Therefore, a slower turn rate of the γ subunit is consistent with partial decoupling of proton flow from ATP synthesis **(Hypothesis 2)**.

(Choices A and C) Hypothesis 1 posits that the mutation results in functional Complex V, but fewer units are available. If L156P Complex V were fully functional but fewer complexes were present, the *total ATP synthesis rate* would slow but the *rotation rate of each individual Complex V unit* would be unaffected.

(Choice D) To support a hypothesis, the data must show specific evidence in favor of that hypothesis. Hypothesis 3 proposes that the pathological effects of the mutation stem from oxidative stress leading to apoptosis. Oxidative stress levels *were not measured* in this experiment, and *apoptosis cannot be monitored in isolated mitochondria*, so the data neither support nor refute the hypothesis that oxidative stress or apoptosis are increased in L156P cells.

Educational objective:

When one chemical process provides the energy for another, the processes are said to be coupled. Decoupling two otherwise coupled processes causes the energy-providing step to continue without driving the dependent process.

Time spent: 0 sec	**QID**:401505
Subject: Biochemistry	**System:** Metabolic Reactions